

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk:

Product Description:

Cat No. :

Synonyms

Molecular Formula

**Bis(2-methoxyethyl) Ether (Certified)**
**Bis(2-methoxyethyl) Ether (Certified)**

O1471-4; O1471-500

Diethylene glycol dimethyl ether; Diglyme; 2-Methoxyethyl ether

C6H14O3

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**E-mail address**

Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888

 CHEMTREC Malaysia **1-800-815-308** (Malay)

 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                       |                      |
|-----------------------|----------------------|
| Flammable liquids     | Category 3 (H226)    |
| Reproductive Toxicity | Category 1B (H360FD) |

**Label Elements**


Signal Word

Danger

**Hazard Statements**

H226 - Flammable liquid and vapor

H360FD - May damage fertility. May damage the unborn child

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## Precautionary Statements

### Prevention

- P201 - Obtain special instructions before use
- P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
- P202 - Do not handle until all safety precautions have been read and understood
- P233 - Keep container tightly closed
- P240 - Ground and bond container and receiving equipment
- P242 - Use non-sparking tools
- P243 - Take action to prevent static discharges
- P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

- P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
- P308 + P313 - IF exposed or concerned: Get medical advice/attention
- P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

### Storage

- P403 + P235 - Store in a well-ventilated place. Keep cool

### Disposal

- P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

- EUH019 - May form explosive peroxides
- This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                        | CAS No   | Weight % |
|----------------------------------|----------|----------|
| Diethylene glycol dimethyl ether | 111-96-6 | > 99     |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | If symptoms persist, call a physician.                                                                                                           |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| Ingestion                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

### Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically. Symptoms may be delayed.

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## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Flammable. Risk of ignition. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. May form explosive peroxides on prolonged storage. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Flammables area. Store under an inert atmosphere. Protect from moisture. Containers should be dated when opened and tested periodically for the presence of peroxides.

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## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                        | European Union | The United Kingdom | Germany                                                                                                                                                                                                                                                            |
|----------------------------------|----------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Diethylene glycol dimethyl ether |                |                    | TWA: 1 ppm (8 Stunden). AGW - exposure factor 8<br>TWA: 5.56 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8<br>TWA: 1 ppm (8 Stunden). MAK<br>TWA: 5.6 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 8 ppm<br>Höhepunkt: 44.8 mg/m <sup>3</sup><br>Haut |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles)

#### Hand Protection

Protective gloves

#### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

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## Information on basic physical and chemical properties

|                                         |                            |                                        |
|-----------------------------------------|----------------------------|----------------------------------------|
| Appearance                              | Clear                      |                                        |
| Physical State                          | Liquid                     |                                        |
| Odor                                    | Petroleum distillates      |                                        |
| Odor Threshold                          | No data available          |                                        |
| pH                                      |                            |                                        |
| Melting Point/Range                     | -64 °C / -83.2 °F          |                                        |
| Softening Point                         | No data available          |                                        |
| Boiling Point/Range                     | 162 °C / 323.6 °F          | @ 760 mmHg                             |
| Flash Point                             | 55 °C / 131 °F             | Method - No information available      |
| Evaporation Rate                        | 0.36 (Butyl Acetate = 1.0) |                                        |
| Flammability (solid,gas)                | Not applicable             | Liquid                                 |
| Explosion Limits                        | No data available          |                                        |
| Vapor Pressure                          | 3.0 mmHg @ 25 °C           |                                        |
| Vapor Density                           | 4.62 (Air = 1.0)           | (Air = 1.0)                            |
| Specific Gravity / Density              | .9370                      |                                        |
| Bulk Density                            | Not applicable             | Liquid                                 |
| Water Solubility                        | Soluble                    |                                        |
| Solubility in other solvents            | No information available   |                                        |
| Partition Coefficient (n-octanol/water) |                            |                                        |
| Component                               | log Pow                    |                                        |
| Diethylene glycol dimethyl ether        | -0.36                      |                                        |
| Autoignition Temperature                | 170 - °C / 338 - °F        |                                        |
| Decomposition Temperature               | No data available          |                                        |
| Viscosity                               | No data available          |                                        |
| Explosive Properties                    |                            | explosive air/vapour mixtures possible |
| Oxidizing Properties                    | No information available   |                                        |
| Molecular Formula                       | C6H14O3                    |                                        |
| Molecular Weight                        | 134.18                     |                                        |
| VOC Content(%)                          | 99                         |                                        |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

Yes.

### Chemical Stability

Stable under normal conditions. May form explosive peroxides. Hygroscopic.

### Possibility of Hazardous Reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

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## Conditions to Avoid

Incompatible products. Heat, flames and sparks. Exposure to moisture. Extremes of temperature and direct sunlight. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition.

## Incompatible Materials

Strong oxidizing agents. Strong acids. Isocyanates.

## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

No data available

Inhalation

No data available

| Component                        | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation                            |
|----------------------------------|---------------------------|---------------------------|--------------------------------------------|
| Diethylene glycol dimethyl ether | LD50 = 7500 mg/kg ( Rat ) | LD50 > 6900 mg/kg ( Rat ) | LC50 > 11000 mg/m <sup>3</sup> ( Rat ) 7 h |

##### (b) skin corrosion/irritation;

No data available

##### (c) serious eye damage/irritation;

No data available

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

##### (e) germ cell mutagenicity;

No data available

##### (f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

##### (g) reproductive toxicity;

Reproductive Effects

Category 1B

Developmental Effects

May impair fertility.

May cause harm to the unborn child.

##### (h) STOT-single exposure;

No data available

##### (i) STOT-repeated exposure;

No data available

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|                                                   |                                                                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Target Organs</b>                              | No information available.                                                                                                           |
| <b>(j) aspiration hazard;</b>                     | No data available                                                                                                                   |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                      |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.                                                |
| <b>Endocrine Disrupting Properties</b>            | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

| Component                        | Freshwater Fish                                                                                                        | Water Flea | Freshwater Algae | Microtox                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------|------------|------------------|-------------------------------|
| Diethylene glycol dimethyl ether | Rainbow trout: LC50 = 9845 mg/L/96h<br>Fathaed Minnow: LC50 = 8569 mg/L 96h<br>Bluegill/Sunfish: LC50 = 10928 mg/L 96h |            |                  | Daphnia: EC50 = 5868 mg/L 96h |

### Persistence and degradability

| <b>Persistence</b>                                  | Persistence is unlikely. |
|-----------------------------------------------------|--------------------------|
| Component                                           | Degradability            |
| Diethylene glycol dimethyl ether<br>111-96-6 (> 99) | >99% 36d (OECD 302B)     |

### Bioaccumulative potential

|                                  |                |                                      |
|----------------------------------|----------------|--------------------------------------|
| <b>Component</b>                 | <b>log Pow</b> | <b>Bioconcentration factor (BCF)</b> |
| Diethylene glycol dimethyl ether | -0.36          | No data available                    |

### Mobility in soil

The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

#### Other Information

Waste codes should be assigned by the user based on the application for which the product was used Do not flush to sewer Can be landfilled or incinerated, when in compliance with

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local regulations

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN3271  
Hazard Class 3  
Packing Group III  
Proper Shipping Name ETHERS, N.O.S. 1,2-Dimethoxyethane

### Road and Rail Transport

UN-No UN3271  
Hazard Class 3  
Packing Group III  
Proper Shipping Name ETHERS, N.O.S. 1,2-Dimethoxyethane

### IATA

UN-No UN3271  
Hazard Class 3  
Packing Group III  
Proper Shipping Name ETHERS, N.O.S. 1,2-Dimethoxyethane

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component                        | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|----------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Diethylene glycol dimethyl ether | 203-924-4 | X    | X   | X     | X    | X    | X     | X    | KE-27705 |

### National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals



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**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

24-Mar-2025

Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**